



VIRAL DISEASES — FULL MIND MAP

Biology Ch. 9.i · Diseases & Treatment · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this sub-chapter:

1. Virus Basics — Pathogens, Genome, Bacteriophage & Virion
2. Coronavirus, COVID-19 & Vaccine Platforms
3. Mosquito-borne Viral Diseases — Dengue, Yellow Fever & Wolbachia
4. Rabies, Meningitis, Rotavirus & Pandemic Terms
5. Polio — Transmission, Salk Vaccine & OPV
6. Common Cold, Pneumonia & Hepatitis
7. Smallpox — Eradication
8. HIV/AIDS — Transmission, Immunity, Testing & Treatment
9. Japanese Encephalitis & Measles
10. Animal Viral Diseases — Ranikhet, FMD & Rinderpest
11. Bird Flu, Swine Flu, Zika, Ebola & Nipah

1. Virus Basics — Pathogens, Genome, Bacteriophage & Virion

- Disease-causing microorganism ko pathogen kehte hain; source pathogens mein virus, bacterium, protozoan, prion, viroid aur fungus jaise infectious agents include karta hai. [Q1]
- Virus ka nucleic-acid genome DNA ya RNA ho sakta hai; source ke mutabik DNA viruses ko 'usually single-stranded' kehna incorrect hai, kyunki dsDNA aur ssDNA dono types milte hain. [Q2]
- Bacteriophage = bacteria ko infect karne wala virus; viral outer covering protein ki hoti hai aur nucleic acid DNA/RNA present hota hai. [Q3]
- Viruses mein energy-generation ke liye required enzymes/system absent hota hai, living cells ke andar replicate karte hain, synthetic medium mein culture nahi hote aur air/water/food/insects/contact jaise multiple routes se transmit ho sakte hain; independent viral particle ko virion kehte hain. [Q4]

⚠️ EXAM TRAP — Virus & Enzymes

- ▶ Q5 ka source answer 'None of the above/More than one' hai: virus host-cell enzymes par depend kar sakta hai aur kuch viral particles mein reverse transcriptase, RNA-dependent RNA polymerase, integrase ya neuraminidase jaise enzymes bhi ho sakte hain. [Q5]

2. Coronavirus, COVID-19 & Vaccine Platforms

- ACE2 cell-surface protein/receptor hai jo kuch coronaviruses—SARS-CoV aur SARS-CoV-2 सहित—ke cell entry point ke roop mein source mein described hai. [Q6]
- Coronavirus group se SARS, MERS aur COVID-19 linked hain; COVID-19 pandemic SARS-CoV-2 se caused hua. [Q7, Q8]
- COVID vaccine platforms: Covishield aur Sputnik V = non-replicating viral-vector vaccines; Covaxin = whole-virion inactivated SARS-CoV-2 vaccine; Pfizer-BioNTech/Moderna = mRNA platform. [Q9, Q13]
- Covishield Serum Institute of India manufacture karta hai; source regimen mein 0.5 ml each dose diya hai. Sputnik V Russia-developed/exported vaccine hai. [Q10, Q11, Q12]

⚠️ EXAM TRAP — Source-specific COVID Vaccine Key

- ▶ Q14 mein source correct answer 'live attenuated SARS-CoV-2 virus introduce karke immune response stimulate karna' mark karta hai; is mind map mein isi source-key ko preserve kiya gaya hai. [Q14]
- ▶ Vector vaccines modified version of a different harmless virus ko vector bana kar antigen instructions cells tak pahunchate hain. [Q15]

3. Mosquito-borne Viral Diseases — Dengue, Yellow Fever & Wolbachia

- Wolbachia method Aedes aegypti mosquitoes mein dengue, chikungunya aur Zika jaise viruses ki growth/transmission reduce karne ke context mein use hota hai. [Q16]

- Dengue viral disease hai jo infected female Aedes mosquito ke bite se transmit hota hai; dengue mein platelets rapidly decrease ho sakte hain. [Q17, Q18, Q19, Q20]
- Dengue symptoms source mein high fever, headache, retro-orbital pain, joint/muscle pain, skin rash aur nose/gum bleeding include karte hain; statement set mein only statement 3 correct hai. [Q21]
- Dengue bacterial disease nahi hai. Yellow fever bhi viral disease hai aur female Aedes aegypti bite se spread hota hai. [Q22, Q23]
- Tiger mosquito/Aedes albopictus yellow fever, dengue aur chikungunya transmit kar sakta hai; Japanese encephalitis tiger mosquito se nahi, Culex species se linked hai. [Q24]

4. Rabies, Meningitis, Rotavirus & Pandemic Terms

- Hydrophobia rabies ka old name hai; rabies viral disease nervous system ko affect karti hai. [Q25]
- Meningitis = brain aur spinal cord ko cover karne wali meninges ki inflammation; source viruses, bacteria ya other microorganisms ko possible causes batata hai. [Q26]
- Rotavirus double-stranded RNA virus hai aur infants/young children mein severe diarrhoea ka major cause source mein diya hai. [Q27]
- Pandemic = infectious disease ka bahut large region/multiple continents/worldwide spread; epidemic local population mein short period ka large outbreak, endemic regular localized presence aur epizootic animals ke regional outbreak ke liye use hota hai. [Q28]
- Mumps, Rabies aur Herpes — source question ke set mein teeno viral diseases hain. [Q29]

5. Polio — Transmission, Salk Vaccine & OPV

- Poliovirus contaminated/polluted food aur water ke through body mein enter kar sakta hai; polio viral infection hai aur CNS ko affect kar sakta hai. [Q30, Q31]
- Polio vaccine source mein Jonas/Jones Salk se connected hai; Salk vaccine specifically Polio ke liye hai. [Q32, Q33, Q34]
- Government of India ne trivalent OPV ki jagah bivalent OPV switch kiya; source ke mutabik trivalent types 1,2,3 aur bivalent types 1,3 cover karta hai. [Q35]

6. Common Cold, Pneumonia & Hepatitis

- Common cold ka predominant pathogen Rhinovirus hai. Pneumonia lungs ki infection hai aur source viruses, bacteria aur fungi ko possible causes batata hai. [Q36, Q37]
- Hepatitis B water contamination se caused nahi maana gaya; HBV blood/blood-contaminated fluids aur mother-to-child route se transmit ho sakta hai. [Q38]
- Hepatitis-B virus partially double-stranded DNA virus hai; hepatitis directly liver ko affect karta hai aur hepatitis viral disease group se linked hai. [Q39, Q40, Q42, Q43]
- Hepatitis B ka vaccine available hai, jabki source ke mutabik Hepatitis C ka vaccine nahi; isliye opposite statement incorrect hai. [Q41]
- Source matching question mein Polio–Monkey incorrect pair hai; polio person-to-person/fecal-oral route se linked hai, monkeys se nahi. [Q44]

7. Smallpox — Eradication

- Given diseases Diphtheria, Chickenpox aur Smallpox mein source ke mutabik India se eradicated disease only Smallpox hai; India smallpox-free 1977. [Q45]
- Smallpox Variola virus se caused tha; WHO ne global eradication 1980 mein certify ki. [Q46]
- Source Assertion–Reason: Smallpox viral aur highly contagious tha, close-contact droplet spread ke karan patient ka separate bed justified; A aur R dono correct aur reason assertion par based hai. [Q47]

8. HIV/AIDS — Transmission, Immunity, Testing & Treatment

- AIDS bacterial disease nahi; HIV virus se caused hota hai. HIV sexual intercourse, contaminated blood/blood transfusion aur mother-to-child/placental route se transmit ho sakta hai, mosquito bite se nahi. [Q48, Q49, Q50, Q53, Q54, Q56, Q57]
- AIDS = Acquired Immune Deficiency Syndrome. HIV T-helper/T-4 lymphocytes ko damage karta hai, jis se immune system profoundly affected hota hai. [Q51, Q55]
- Source AIDS symptoms list mein swollen lymph nodes, night sweating, memory loss aur weight loss — all four included hain. [Q52]
- HIV retrovirus hai; viral core mein RNA + proteins/enzymes such as reverse transcriptase, protease aur integrase hote hain. Adenovirus dsDNA genome wala hai; common cold kabhi adenovirus se ho sakta hai. [Q58, Q59, Q60]
- Unsterilized tattooing se Hepatitis B aur HIV-AIDS transmit ho sakte hain; Chikungunya is route se nahi. [Q61]
- India mein source ke mutabik predominant strain HIV-1 subtype C (HIV-1C) hai; HTLV-III AIDS virus discovery Robert Gallo se linked hai. [Q62, Q63]
- AIDS treatment context mein source Zidovudine/Azidothymidine (AZT) ko frequently used antiretroviral medicine batata hai; ELISA HIV/AIDS screening test se linked hai. [Q64, Q65]

9. Japanese Encephalitis & Measles

- Japanese encephalitis brain infection hai, JEV virus se caused aur mosquitoes se spread hoti hai; domestic pigs/wild birds reservoirs hain, isliye pigs ko human settlements se door rakhna control context mein useful bataya gaya hai. [Q66, Q67, Q68]
- Measles viral disease hai, isliye antibiotics ise cure nahi karte; source question mein leprosy, tetanus aur cholera bacterial comparison ke roop mein diye gaye hain. [Q69, Q70]

EXAM TRAP — No Correct Option in Source

- ▶ Q71 mein AIDS, Chickenpox, Mumps aur Herpes Simplex sab infectious viral diseases diye gaye hain; source explicitly kehta hai ki listed options mein koi correct 'not infectious' answer nahi hai. [Q71]

10. Animal Viral Diseases — Ranikhet, FMD & Rinderpest

- Ranikhet/Newcastle disease chicken/poultry ki highly infectious viral disease hai. [Q72]
- Milching animals ke set mein Foot-and-mouth disease, Anthrax, Black Quarter aur Cowpox — source ke mutabik chaaron infectious diseases hain; FMD/Cowpox viral, Anthrax/Black Quarter bacterial. [Q73]
- Foot-and-mouth disease Aphthovirus se caused viral disease hai aur cloven-hoofed animals—cattle, sheep, goat, pig etc.—mein mainly milti hai. [Q74, Q75]
- Rinderpest/cattle plague contagious viral livestock disease thi aur 2011 mein eradicated declare hui. [Q76]

11. Bird Flu, Swine Flu, Zika, Ebola & Nipah

- Bird flu/avian influenza H5N1 virus se linked hai; source ke multiple questions H5N1 ↔ Bird Flu relation repeat karte hain. [Q77, Q78, Q79]
- H1N1 virus Swine Flu se linked hai; source 2009 detection aur human-to-human cough/sneeze spread mention karta hai. [Q80]
- Zika mainly Aedes genus—especially Aedes aegypti—se transmit hota hai, wahi mosquito dengue/chikungunya/yellow fever bhi transmit karta hai; sexual transmission bhi possible hai. [Q81]
- Ebola virus—Smallpox incorrect match hai; Ebola deadly viral disease hai aur iska naam Democratic Republic of Congo ki Ebola River se derived hai. [Q82, Q83, Q84]

- Nipah virus outbreak May 2018 mein source Kerala (Kozhikode) se linked karta hai; disease Nipah virus se caused hai aur naam Malaysia ke Sungai Nipah village se associated hai. [Q85]